Math 1200 Review Session 3



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Office hours: Fridays 4-5pm, WLH 015

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$x =$ the length of each of the two ends,

$y =$ the length of the side opposite the barn.

$$A = A(x, y) = xy$$

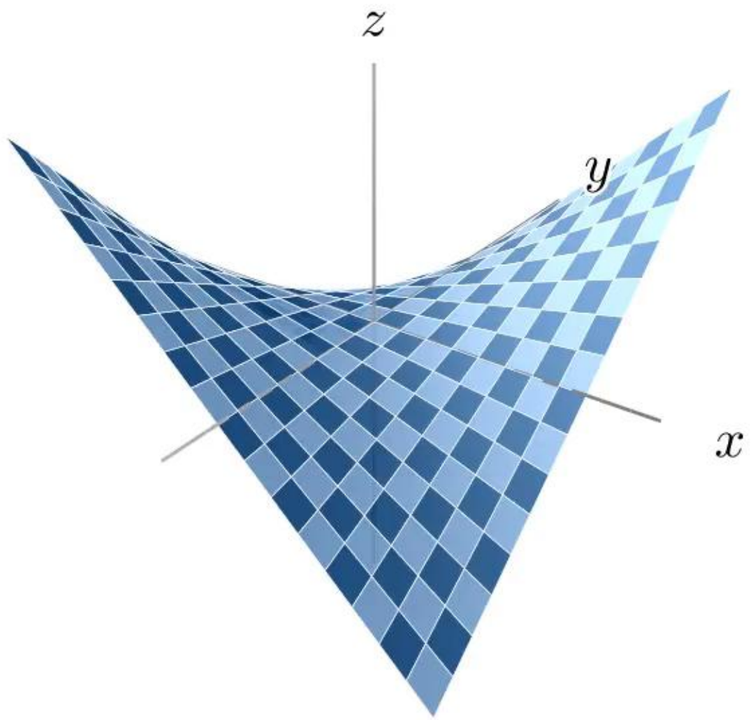
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Can you draw this function of two variables?

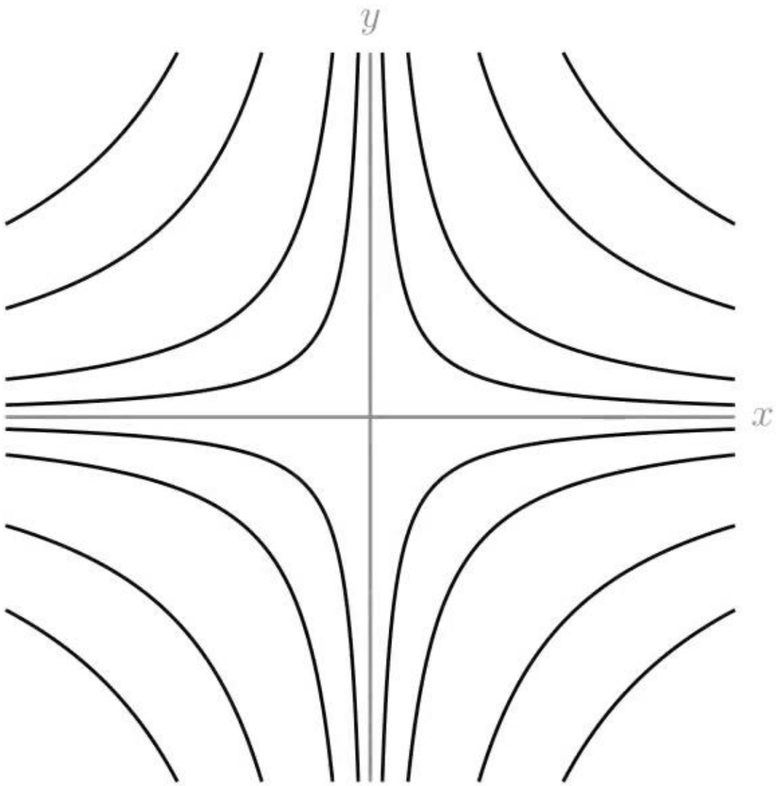
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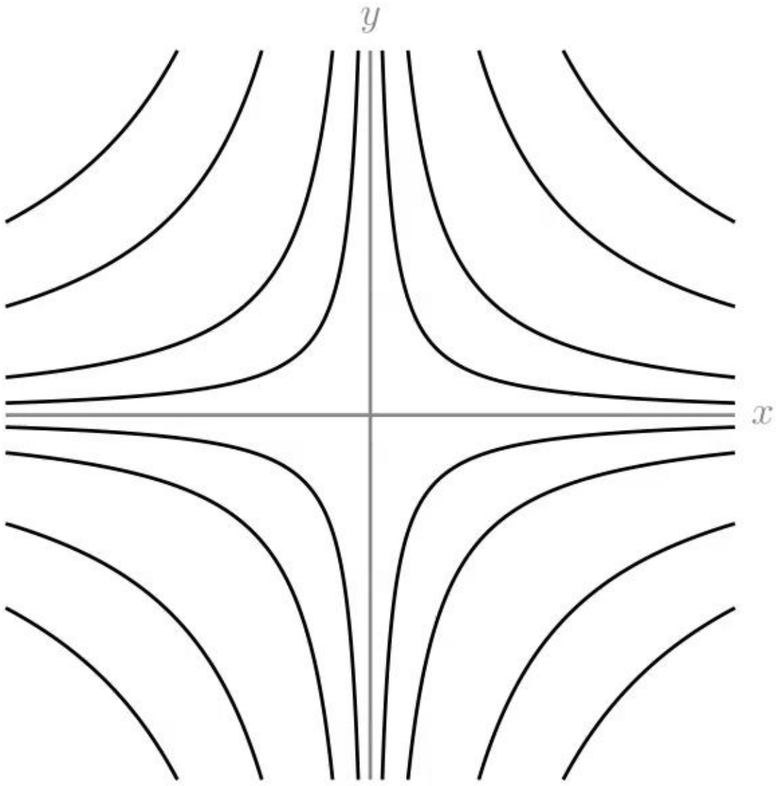
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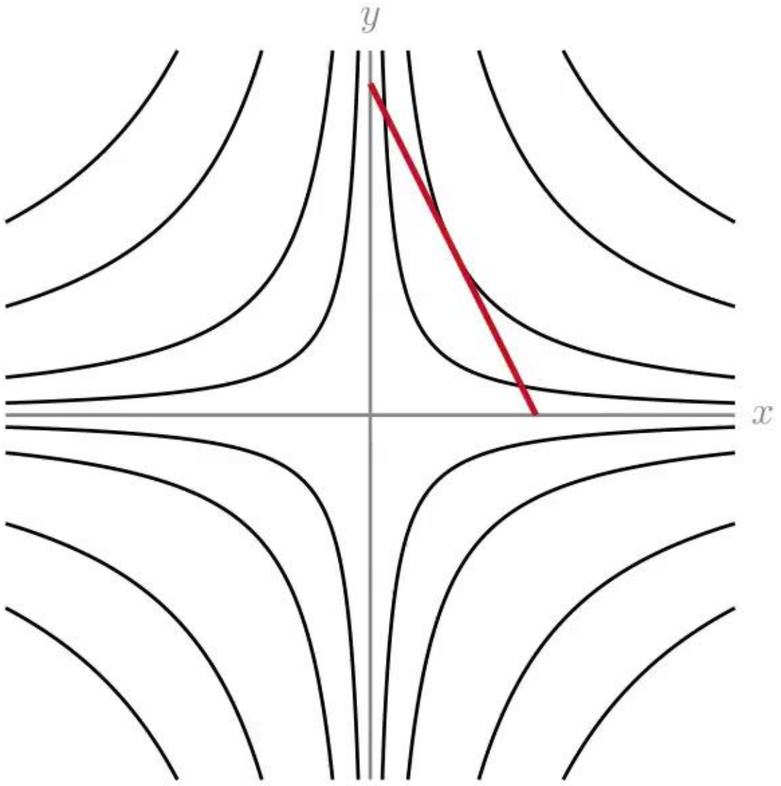
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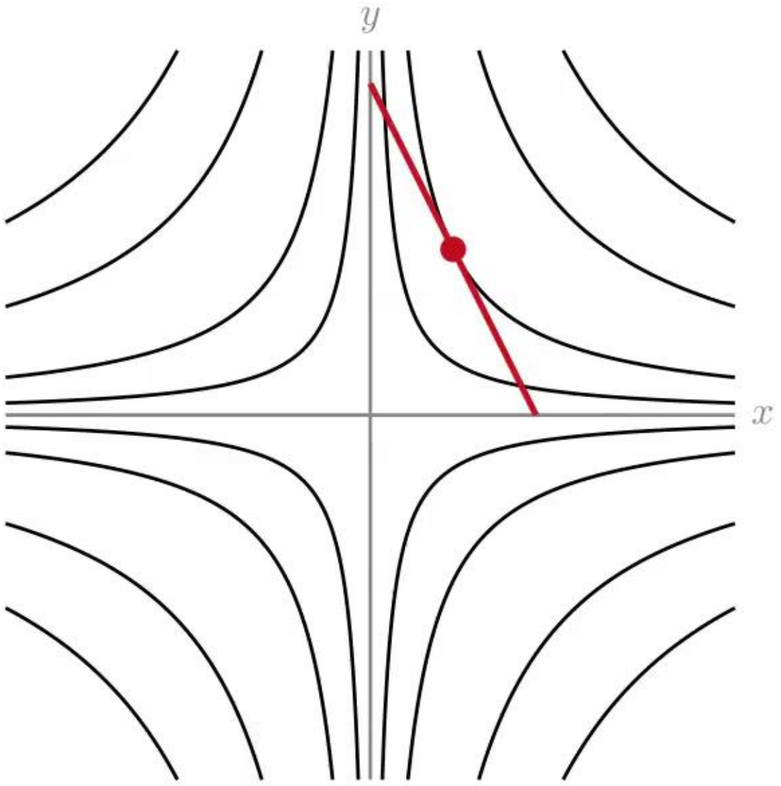
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$$x = 25 \text{ ft}, \quad y = 50 \text{ ft}, \quad A = 1250 \text{ ft}^2$$

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$(1, 1)$, distance $\sqrt{5}$

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$$r = \sqrt[3]{250/\pi} \approx 4.30 \text{ cm}, \quad h = 2r \approx 8.60 \text{ cm}$$

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$x = 100$: all of it into the circle, $\quad A = \dfrac{2500}{\pi} \approx 796 \text{ cm}^2$

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$$(1,\, 1), \quad \text{distance } \frac{4}{\sqrt{5}}$$

Aside: volumes

A glass is a perfect cone.

Held upside down, it is 2/3 full by height.

How full would it be right-side-up?



Water is poured into an upside-down cone (point at the bottom) of radius 5 ft and height 10 ft at a rate of 2 ft^3/min.
How fast is the water level rising when the water is 6 ft deep?

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$$\frac{dh}{dt} = \frac{2}{9\pi}\ \text{ft/min} \approx 0.071\ \text{ft/min}$$

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$$S(x, y) = xy + \frac{64}{x} + \frac{64}{y}$$

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$$x = y = 4 \text{ in}, \quad z = 2 \text{ in}, \quad S = 48 \text{ in}^2$$